DECENT COACHING CLASSES

VIII CBSE

Mind Your Math - II

Subject: Mathematics — Understanding Quadrilaterals (Parallelogram & Rhombus)

Time: 1 Hour **Total Marks: 30**

Note: (i) All questions are compulsory. (ii) Section A carries 1 mark each, Section B carries 2 marks each, Section C carries 3 marks each and Section D carries 6 marks. (iii) Draw neat figures wherever required.

Section A — Choose the correct alternative. [5]

1. In a parallelogram, opposite angles are *(1)*

(a) equal (b) supplementary (c) complementary (d) unequal

2. The diagonals of a rhombus intersect each other at *(1)*

(a) 60° (b) 90° (c) 120° (d) 180°

3. In parallelogram ABCD, if ∠A = 70°, then ∠B = *(1)*

(a) 70° (b) 110° (c) 90° (d) 180°

4. Which property is true for a rhombus but not necessarily for every parallelogram? *(1)*

(a) Opposite sides are equal (b) Diagonals bisect each other (c) All four sides are equal (d) Opposite angles are equal

5. The diagonals of a parallelogram *(1)*

(a) are equal (b) bisect each other (c) are perpendicular (d) bisect the angles

Section B — Answer the following questions. [10]

1. In parallelogram PQRS, PQ = 8 cm. Find the length of RS, giving reason. *(2)*

2. Two adjacent angles of a parallelogram are in the ratio 4 : 5. Find the measures of all four angles. *(2)*

3. In rhombus ABCD, diagonal AC = 16 cm and diagonal BD = 12 cm. Find the length of a side of the rhombus. *(2)*

4. State two properties of a parallelogram that also hold for a rhombus, and one property that holds for a rhombus but not for every parallelogram. *(2)*

5. In parallelogram ABCD, diagonals AC and BD intersect at O. If AO = 5 cm, find the length of OC. Give reason. *(2)*

Section C — Answer the following questions. [9]

1. ABCD is a parallelogram in which ∠A = $(2x + 15)°$ and ∠C = $(3x − 25)°$. Find the value of x and hence find the measures of all four angles of the parallelogram. *(3)*

2. In rhombus PQRS, diagonal PR = 24 cm and side PQ = 13 cm. Find the length of diagonal QS. *(3)*

3. The lengths of the diagonals of a rhombus are in the ratio 3 : 4 and its perimeter is 40 cm. Find the length of each diagonal. *(3)*

Section D — Answer the following question. [6]

1. Prove that in a parallelogram, opposite sides are equal. *(6)*

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VIII CBSE | Mind Your Math - II — ANSWER KEY

Section A Answers

1. (a) equal 2. (b) 90° 3. (b) 110° 4. (c) All four sides are equal 5. (b) bisect each other

Section B Answers

1. In a parallelogram, opposite sides are equal. ∴ RS = PQ = 8 cm

2. Adjacent angles of a parallelogram are supplementary. Let the angles be $4x$ and $5x$. ∴ $4x + 5x = 180° \Rightarrow 9x = 180° \Rightarrow x = 20°$. ∴ Angles are 80°, 100°, 80°, 100° (opposite angles equal)

3. Diagonals of a rhombus bisect each other at right angles. Half-diagonals = 8 cm and 6 cm. Side = $\sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$ cm

4. Properties also true for a rhombus: (i) opposite sides are parallel and equal, (ii) diagonals bisect each other. Property true only for a rhombus: all four sides are equal (diagonals also bisect each other at right angles).

5. Diagonals of a parallelogram bisect each other. ∴ OC = AO = 5 cm

Section C Answers

1. In a parallelogram, opposite angles are equal. ∴ $\angle A = \angle C \Rightarrow 2x + 15 = 3x - 25 \Rightarrow x = 40$. ∴ $\angle A = \angle C = 2(40) + 15 = 95°$. Since adjacent angles are supplementary, $\angle B = \angle D = 180° - 95° = 85°$

2. Diagonals of a rhombus bisect each other at right angles. Half of PR = 12 cm. In right triangle formed, $PQ^2 = 12^2 + (\text{half of } QS)^2 \Rightarrow 13^2 = 144 + (\text{half } QS)^2 \Rightarrow (\text{half } QS)^2 = 169 - 144 = 25 \Rightarrow$ half QS = 5 cm $\Rightarrow$ QS = 10 cm

3. Side of rhombus = Perimeter ÷ 4 = 40 ÷ 4 = 10 cm. Let diagonals = $3x$ and $4x$, so half-diagonals = $1.5x$ and $2x$. By Pythagoras: $(1.5x)^2 + (2x)^2 = 10^2 \Rightarrow 2.25x^2 + 4x^2 = 100 \Rightarrow 6.25x^2 = 100 \Rightarrow x^2 = 16 \Rightarrow x = 4$. ∴ Diagonals = 3(4) = 12 cm and 4(4) = 16 cm

Section D Answer

1. Given: ABCD is a parallelogram with AB || CD and AD || BC. To prove: AB = CD and AD = BC. Construction: Join diagonal AC. Proof: In △ABC and △CDA, ∠BAC = ∠DCA (alternate angles, AB || CD) and ∠BCA = ∠DAC (alternate angles, BC || AD), and AC = CA (common side). ∴ △ABC ≅ △CDA (by ASA congruence rule). ∴ AB = CD and BC = AD (corresponding sides of congruent triangles). Hence, opposite sides of a parallelogram are equal. (Hence proved.)